EXP 1	A PYTHON PROGRAM TO ASOLVE A LPP - CONSTRAINT OPTIMIZATION

AIM:

To solve a Linear Programming Problem (LPP) for constraint optimization using Python.

PROCEDURE:

1. Install Required Libraries:

o Make sure you have numpy and scipy installed. You can install them using pip if

not already installed.

bash

Copy code

pip install numpy scipy

2. Define the Problem:

o Identify the objective function to maximize or minimize.

o Determine the constraints (equality and inequality).

3. Set Up the Problem in Python:

o Use the scipy.optimize.linprog function to set up and solve the LPP.

4. Run the Code:

o Execute the Python code to get the optimized result.

5. Analyze the Output:

o Interpret the results and validate the solution against the problem constraints.

PROBLEM 1:

CODE:

Code to solve a Linear Programming Problem using Python.

Formulate and solve the following Linear Programming Problem using Python:

Maximize

$$Z = 3x_1 + 2x_2$$

Subject to:

- $2x_1 + x_2 \leq 20$
- $4x_1 - 5x_2 \leq 10$
- $x_1, x_2 \geq 0$

```
import numpy as np
from scipy.optimize import linprog
# Define the coefficients of the objective function
# For example, maximize: z = 3x1 + 2x2
c = [-3, -2]

# Define the coefficients of the inequality constraints
# For example:
# 2x1 + x2 <= 20
# 4x1 - 5x2 <= 10
# x1, x2 >= 0 (non-negativity constraints)
A = [
[2, 1],
[4, -5]
]
```

```
# Define the right-hand side of the inequality constraints
b = [20, 10]
# Define bounds for variables (non-negativity constraints)
x_bounds = (0, None)
y_bounds = (0, None)
# Solve the linear programming problem
result = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds],
method='highs')
# Print the result
print('Optimal value:', -result.fun)
print('Values of x:', result.x)
```

EXPECTED OUTPUT:

Optimal value: 14.0

Values of x: [10. 0.]

PROBLEM 2:

Formulate and solve the following Linear Programming Problem using constraint optimization:

Minimize

$$\boldsymbol{Z = 20x + 10y}$$

Subject to:

$$\boldsymbol{x + 2y \geq -40}$$

$$\boldsymbol{3x + y \geq 30}$$

$$\boldsymbol{4x + 3y \geq 60}$$

$$\boldsymbol{x \geq 0,\ y \geq 0}$$

CODE:

```
import numpy as np
from scipy.optimize import linprog
```

```
c = [20, 10]
A = [
    [-1, -2],
    [-3, -1],
    [-4, -3]
]
b = [40, -30, -60]
x_bounds = (0, None)
y_bounds = (0, None)
result = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds],
method='highs')
print("Optimal value:", result.fun)
print("Values of x:", result.x)
```

EXPECTED OUTPUT:

Optimal value: 240.0

Values of x: [6. 12.]

RESULT: Thus, the Linear Programming Problem (LPP) by Constraint Optimization using Python program has been implemented successfully.